

IDS Forge ■■■ — System Access & Execution Guide

Machine Learning-Based Hybrid Intrusion Detection System for IoT Networks

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1. Quick Start (Running on Your Local Machine)

To launch the IDS Web Dashboard immediately on your machine, open **Command Prompt** or **PowerShell** in the project root folder and run:

```
python -m streamlit run app.py
```

Your browser will automatically open: <http://localhost:8501>

2. Full Installation & Setup Guide (For Supervisors / Reviewers)

Anyone cloning this repository from GitHub can set up and run the system in 3 simple steps:

Step 1: Clone the GitHub Repository

```
git clone https://github.com/lakmal6214/IDS-Forge-Research.git
cd IDS-Forge-Research
```

Step 2: Install Required Dependencies

```
python -m pip install -r requirements.txt
(Or: py -m pip install -r requirements.txt)
```

Step 3: Launch Option A — Interactive Web Dashboard

```
python -m streamlit run app.py
```

Access the interactive UI in browser at <http://localhost:8501>.

Step 4: Launch Option B — Terminal Experimental Pipeline

```
python main.py
```

Executes all 8 experimental stages (data loading, feature selection, signature matching, ML classifier training, zero-day simulation, and plot rendering).

3. Professional Repository Architecture

The codebase is structured following industry-standard Python package design:

Directory / File	Description & Purpose
app.py	Interactive Streamlit Web UI Dashboard entrypoint.
main.py	CLI Experimental Pipeline Orchestrator (Stages 1-8).
src/	Core engine source package (data loader, feature selection, signatures, ML models, hybrid IDS, evaluator, visualizer).
docs/	Complete submission deliverables (Dissertation, Viva Presentation, PDF Report, Technical Guides).
scripts/	Utility generators for Word, PowerPoint, and PDF documentation.
output/	Generated publication figures (Figures 1-8) and CSV benchmark results.

4. Academic Deliverables & Documents Index

All academic artifacts located in docs/:

Document File	Format	Description
Dissertation_IDS_Forge.docx	MS Word	Complete 8,000-10,000 word dissertation.
Viva_Presentation_IDS_Forge.pptx	PPTX	35-slide deck with speaker notes.
Final_Report_IDS_Forge.pdf	PDF	Final submission report PDF.
User_Guide_How_To_Run_IDS_Forge.pdf	PDF	System access & step-by-step execution guide.
Technical_Documentation.md	Markdown	Architecture & edge gateway deployment guide.
Submission_Checklist.md	Markdown	COM4901 submission checklist.